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Multi-modal music techniques for synthesizing high-quality audio waveforms from MIDI data

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A highly effective music synthesizer should deliver high-fidelity audio for a mix of instruments and voices. Current synthesizers often need to choose between specialized models that provide detailed control over specific instruments and flexible waveform models that accommodate a variety of music at the expense of precision. To transcend the existing limitations, this paper introduces MIAO, an avant-garde neural music synthesizer that revolutionizes the domain of interactive and expressive music synthesis by converting MIDI sequences into rich, dynamic audio outputs. Specifically, MIAO can be cultivated through training on diverse transcription datasets that correlate MIDI with audio, thereby deepening its comprehension of MIDI intricacies and elevating its capacity for robust representation learning. This approach allows MIAO to offer precise note-level control over composition and instrumentation, effectively handling a wide spectrum of instruments. We evaluate MIAO's performance by benchmarking it against six datasets: MAESTROv3 (piano), Slakh2100 (synthetic multi-instrument), Cerberus4 (synthetic multi-instrument), Guitarset (guitar), MusicNet (orchestral multi-instrument), and URMP (orchestral multi-instrument), where it sets new performance benchmarks.

Keywords Music synthesizer, MIDI, Multi-modal learning, Audio generation

Instrument music synthesis is a dynamic field within computational musicology that deals with the automated generation of music from multiple instruments[1–3]. A fundamental component of this field involves the use of MIDI (Musical Instrument Digital Interface) music, which refers to a digital music format that stores musical information as a series of electronic messages containing performance data rather than audio waveforms[4]. This area integrates methodologies from computer vision and natural language processing to interpret and replicate the multifaceted nature of orchestral and ensemble music[5,6]. As these technologies have matured, they have greatly enhanced our capacity for audio analysis and the subsequent synthesis of complex musical compositions[7,8]. The realm of multi-instrument music synthesis finds utility in a broad array of applications, including but not limited to, the augmentation of interactive entertainment, the enrichment of virtual reality environments[9,10], the provision of adaptive background scores in film and gaming[11,12], and the customization of therapeutic soundscapes for health and wellness[13–15].

The field of multi-instrument music synthesis presents considerable challenges, primarily due to the inherent compromises between model specialization and versatility[16,17]. Current synthesizers face a pivotal decision: whether to use specialized models that offer precise control over individual instruments, thereby achieving high harmony and accuracy in replicating the unique acoustic properties of those instruments[18,19]. These specialized models excel in capturing the delicate nuances and expressive dynamics essential for defining the timbre and articulation of specific instruments. However, the depth of control these models provide often comes at the cost of flexibility, thereby restricting their use in a broader range of musical genres and the diverse instrument arrangements found in multi-instrument compositions[20,21].

Additionally, flexible waveform models exist at the opposite end of the spectrum, crafted to be indifferent to specific instrument types[22,23]. This design enables them to embrace a wide variety of musical styles and ensembles under a unified framework, catering to the eclectic nature of multi-instrumental compositions[24,25]. While these models excel in versatility, allowing for broad coverage of musical possibilities, they often do so at the cost of precision[26,27]. The challenge here is the dilution of detail, where the intricacies of specific instrument sounds may not be rendered with the same level of accuracy and authenticity as the specialized models[8,28]. This dichotomy between specialization for depth and flexibility for breadth presents a significant hurdle in the pursuit of creating

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a universal music synthesis model that can not only generate a wide variety of music but also maintain the high standards of precision required for professional audio production[29,30].

In order to address the limitations of traditional music synthesis approaches, we introduce the MIAO synthesizer, a novel Multi-mode system that effectively converts **MI**DI inputs into high-fidelity **A**udi**O** outputs. The design of the MIAO synthesizer is focused on producing high-quality audio that accurately represents a wide range of instruments and vocal textures, which are critical for realistic music synthesis. The architecture of the MIAO synthesizer is built around three essential components: the MIDI Encoder, the Audio Decoder, and the Audio Encoder. These elements are strategically implemented to ensure operational efficiency across different phases of the synthesizer's use: the MIDI Encoder and the Audio Decoder are active during both the training and inference phases, while the Audio Encoder is specifically vital during the training phase to enhance audio quality.

Furthermore, this research introduces a supplementary training approach that enhances the performance of the synthesizer beyond its innovative architecture. This scheme employs a dedicated encoder to process and learn from paired MIDI and audio files encompassing a broad spectrum of sounds and instrumental timbres. The effectiveness of the MIAO synthesizer is enhanced through three targeted visual-linguistic objectives: MIDI-audio contrastive learning, MIDI-audio matching, and MIDI-conditioned audio modeling. These objectives are supported by training on diverse datasets that pair MIDI with corresponding audio files, reinforcing the model's ability to learn and adapt. Extensive experimental evaluations have demonstrated that these training strategies significantly improve the MIAO synthesizer's understanding of MIDI sequences and its ability to learn representations. Such advancements enable the synthesizer to exert precise control over musical composition and instrument processing at the note level, effectively managing a large repertoire of instruments. Consequently, the MIAO synthesizer not only stands out as a technological breakthrough in music generation but also serves as a versatile instrument for creative and interactive musical expression, validated by thorough experiments and analyses.

The main contributions in this paper are summarized as follows:

- In the burgeoning field of computational musicology, we propose MIAO, a multi-modal music synthesizer that represents a significant leap forward in MIDI to high-fidelity audio conversion. MIAO is designed with the ambition to close the quality and versatility gap that has long existed in musical synthesis, thereby offering a more nuanced and richly textured musical generation capability.
- A training scheme that complements MIAO has been proposed, integrating three visual-linguistic objectives: MIDI-audio contrastive learning, MIDI-audio matching, and MIDI-conditioned audio modeling. These objectives are meticulously tailored to enhance the model's understanding and replication of musical structure, resulting in a more authentic and expressive synthesis of multi-instrumental compositions.
- In validating MIAO's effectiveness, we benchmarked it across six datasets: MAESTROv3, Slakh2100, Cerberus4, Guitarset, MusicNet, and URMP, encompassing a range of instruments and musical styles. MIAO consistently sets new performance benchmarks, demonstrating its exceptional performance and versatility in multi-instrument music synthesis.

Related work

Neural audio synthesis initially became viable with autoregressive models for raw waveforms, which sequentially predict each audio sample based on all previous ones, establishing a foundation for various complex audio generation tasks[31]. WaveNet[32] is a fully probabilistic, autoregressive model that bases the prediction of each audio sample on all preceding samples. In contrast, SampleRNN[33] utilizes recurrent neural networks across various scales to capture longer-term dependencies in audio waveforms, optimizing for memory efficiency during training. The Generative Adversarial Network (GAN)[34] has been effectively adapted for music generation, facilitating the production of innovative audio waveforms and symbolic compositions that emulate learned musical styles through adversarial training between a generator and a discriminator. WaveGAN[35] diverges from these approaches by generating audio in a single forward pass, focusing on unsupervised synthesis of raw-waveform audio. GANSynth[36] advances this further by producing high-fidelity and locally coherent audio, modeling log magnitudes and instantaneous frequencies in the spectral domain. However, GANSynth, like other GAN-based models, typically concentrates on generating single instruments, notes, or voices at a time[37,38].

Building on the foundations laid by autoregressive and GAN-based models, newer architectures further expand the landscape of neural audio synthesis[39]. Soundstream utilizes Transformers to model the discrete, vector-quantized codes of a foundational waveform autoencoder[40]. Tacotron[41] architectures have shown the efficacy of simple spectrograms for multi-stage audio generation, first creating continuous-valued spectrograms autoregressively, followed by waveform synthesis using a neural vocoder. Denoising Diffusion Probabilistic Models (DDPMs)[42] transform random noise into realistic data through iterative refinement, using a forward process that adds noise and a learned reverse process that removes it. These models outperform GANs in generating high-quality, diverse outputs for audio and image synthesis, making them particularly effective for music generation tasks. MI-DDPM[43] adopts a two-stage training process: it initially uses an autoregressive spectrogram generator alongside a GAN spectrogram inverter and then enhances this with a DDPM spectrogram generator. However, these models require additional focus on spectrograms, along with MIDI inputs, and the DDPM model faces challenges in effectively capturing the contextual representation of waveform autoregression during training[44]. To address these challenges, we introduce an end-to-end neural music synthesizer, optimized for high-quality audio generation across various instruments and voices.

Method

This section introduces MIAO, an advanced multi-modal music synthesizer that marks a substantial advancement in converting MIDI to high-fidelity audio. MIAO is strategically crafted to bridge the longstanding divide in quality and adaptability found in music synthesis, thus providing a more sophisticated and detailed capability for generating musical compositions.

Model architecture

The MIAO consists of three pivotal components: the MIDI Encoder, the audio decoder, and the audio encoder. The MIDI Encoder and audio decoder are consistently operational across both the training and inference phases, while the audio encoder plays a crucial role exclusively during the training phase.

(1) The **MIDI Encoder** is designed with precision to process MIDI data effectively, focusing on refining musical information for accurate audio synthesis. As shown in Fig. 1, assuming the input MIDI data is denoted as x_m, with dimensions of $B \times T \times N$ (representing batch size, time step size, and note/event size respectively), the MIDI Encoder first samples it using deep convolutional layers and reduces it to $\frac{B}{4} \times \frac{T}{4} \times \frac{N}{4}$ dimensions. It then normalizes the data using layer normalization and GELU activation[45], as shown in Eq. 1.

$$data_s = g(LN(dw(x_m)|_\Gamma)) \tag{1}$$

where $data_s$ denotes the standardized data after processed, g represents the GELU activation function, LN represents the layer normalization function, and $dw(*)|_\Gamma$ represents a depthwise convolutional layer with a 3×3 convolutional kernel.

In order to enable the encoder to extract detailed musical nuances crucial for generating high-quality audio, the MIDI Encoder is configured with dual parallel convolutional blocks. Each block is characterized by different convolutional layers with different kernel sizes, followed by batch normalization to enhance the model's generalization ability, as shown in Eq. 2.

$$\begin{aligned} data_{step} &= \begin{cases} data_s \quad , & step = 1, \\ \beta(dw(data_s)|_\Lambda), & step = 2, \\ \beta(dw(data_s)|_\Upsilon), & step = 3. \end{cases} \\ data_m &= \sum_{i=1}^{3} data_i \end{aligned} \tag{2}$$

where $data_m$ represents the dual parallel convolution block output, β represents the batch normalization function. The processing involves two types of 2D depthwise convolutions: $dw|_\Upsilon$ and $dw|_\Lambda$. Specifically, dw_Υ uses a kernel and padding size of 1, while dw_Λ employs a kernel size of 3, with both stride and padding sizes set to 1.

Additionally, the MIDI Encoder is equipped with an encoding attention module to enhance the encoding process, capturing complex sequence dependencies in MIDI data. The implementation procedure is depicted in Eq. 3.

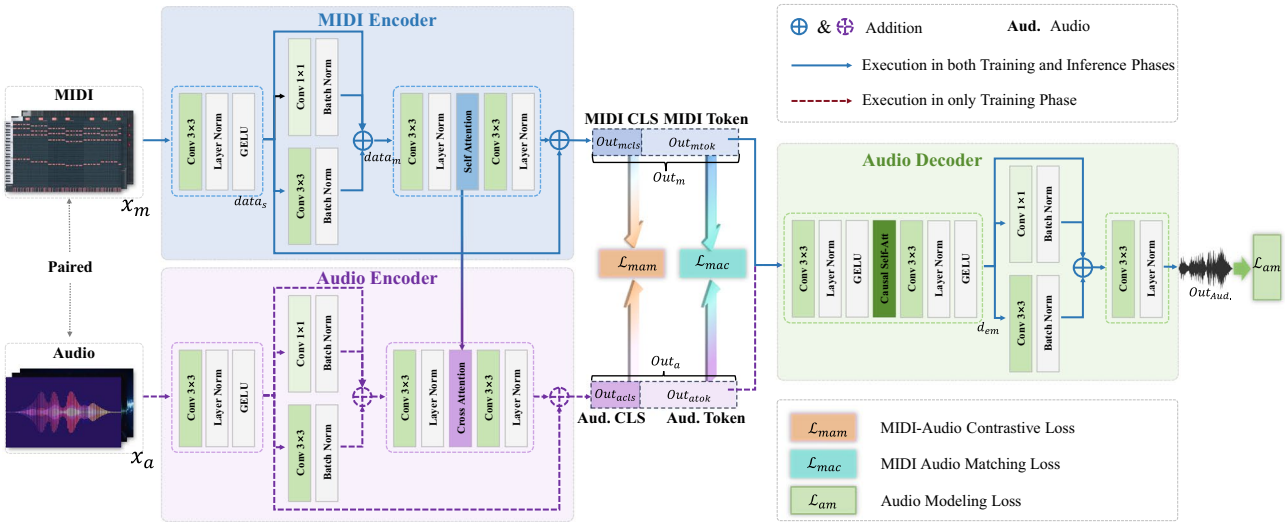

Fig. 1. The overview of the MIAO framework. The MIAO synthesizer architecture features three core components: a MIDI Encoder, an Audio Decoder, and an Audio Encoder, the last of which is utilized exclusively during the training phase. The system is trained on paired MIDI and audio files representing a variety of instruments, employing three distinct learning objectives to optimize performance: MIDI-audio contrastive learning, MIDI-audio matching, and MIDI-conditioned audio modeling.

$$Out_m = dw(LN(SA(LN(dw(data_m)|_\Gamma)))|_\Gamma + data_s. \tag{3}$$

where Out_m represents the final encoding result of the MIDI Encoder, and *SA* represents the self-attention computation mechanism, implemented as shown in Eq. 4[46].

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{4}$$

The tensor $\mathbf{Out_m}$ represents the final output of our MIDI encoder, encoding both structural and semantic information from the input MIDI sequence x_m. This output is partitioned into two components: $\mathbf{Out_{mcls}}$, which serves as a classification (CLS) token capturing global sequence-level features, and $\mathbf{Out_{mtok}}$, comprising MIDI tokens that encode local musical semantics. This architectural division supports parallel optimization of complementary objectives: holistic representation learning through $\mathbf{Out_{mcls}}$ and preservation of fine-grained melodic patterns via $\mathbf{Out_{mtok}}$.

(2) The **Audio Encoder** is specifically designed for integrating audio-specific information x_a and generating the audio encode result Out_a, with a structure similar to that of the MIDI Encoder. It enhances the model's focus on important audio features by incorporating an additional cross-attention layer instead of the self-attention layer used in the MIDI encoder. Additionally, a specialized token, Audio Encoding, is attached to the audio input, with its embedding serving as a multimodal representation combining MIDI and audio data. This is crucial during training and allows for the omission of the audio encoder during inference, simplifying the process.

The audio encoder is designed to capture detailed information from audio sequences. Its components work together to process and refine the musical content, preparing it for high-quality audio translation. The encoder architecture is constructed to capture both spatial and temporal characteristics of the audio, making it a valuable component in the music synthesis pipeline. To facilitate streamlined training and capitalize on multi-task learning, all parameters between the MIDI encoder and audio decoder are identical, except for those within the attention-based layers. This distinction is intentional, as the nuances distinguishing the encoding from decoding processes are most effectively represented within the attention mechanisms. Simultaneously, similar to the MIDI Encoder, the output of this Encoder will be divided into Out_{atok} and Out_{acls} for loss calculation.

(3) The **Audio Decoder** is responsible for converting encoded data *em* into high-fidelity audio. This is achieved through two carefully designed parts of the network layers and function combinations. The first part combines deep convolutional layers and channel self-attention computation to parse the features of the encoded data, as shown in Eq. 5.

$$de_m = g(LN(CSA(g(LN(dw(em)|_\Gamma))))) \tag{5}$$

where de_m denotes the output of this compute phase, *CSA* denotes the channel self-attention computation. Moreover, the second part uses a combination of convolutional layers with different kernel sizes to refine the audio data systematically, as shown in Eq. 6. This configuration not only helps address the gradient vanishing problem but also enriches the decoder's output by retaining important information that may be lost in deep network structures.

$$\begin{aligned} de_{step} &= \begin{cases} de_m \quad , & step = 1, \\ \beta(dw(de_m)|_\Lambda), & step = 2, \\ \beta(dw(de_m)|_\Upsilon), & step = 3. \end{cases} \\ Out_{Aud.} &= LN\left(dw\left(\sum_{i=1}^{3} de_i)|_\Gamma\right)\right) \end{aligned} \tag{6}$$

where $Out_{Aud.}$ denotes the final generation audio result.

Such a design indicates a concerted effort to capture a full spectrum of audio features, from basic waveforms to complex overtones, thus equipping the Audio Decoder to reconstruct a rich and textured audio output from the abstract representations received from the earlier stages of the model. This architectural finesse positions the Audio Decoder as a key component in the synthesis pipeline, pivotal for achieving the end goal of generating synthesized music that is both rich in detail and high in quality. The details are illustrated in Algorithm 1.

Input: MIDI x_m and paired Audio x_a;
Output: Generate Audio *Audio*;

Calculate the **MIDI Encoder** calculation result Out_m:

$$data_s = g(LN(dw(x_m)|_\Gamma))$$

$$data_m = data_s + \beta(dw(data_s)|_\Lambda) + \beta(dw(data_s)|_\Upsilon)$$

$$Out_m = dw(LN(SA(LN(dw(data_m)|_\Gamma)))|_\Gamma$$

$$+data_s$$

Calculate the **Audio Encoder** calculation result Out_a:

$$data_s = g(LN(dw(x_a)|_\Gamma))$$

$$data_m = data_s + \beta(dw(data_s)|_\Lambda) + \beta(dw(data_s)|_\Upsilon)$$

$$Out_a = dw(LN(SA(LN(dw(data_m)|_\Gamma)))|_\Gamma$$

$$+data_s$$

Take **token** and **cls** of Out_m and Out_a to calculate loss L_1:

$$L_1 = L_{MAC}(Out_{mtok}, Out_{atok})$$

$$+L_{MAM}(Out_{mcls}, Out_{acls})$$

Calculate the Generate Audio *Audio* of **Audio Decoder**:

$$em = Out_a + Out_m$$

$$de_m = g(LN(CSA(g(LN(dw(em)|_\Gamma))))))$$

$$Out_{Aud.} = LN(dw(de_m + \beta(dw(data_s)|_\Lambda)|_\Gamma)$$

$$+\beta(dw(data_s)|_\Upsilon)))$$

Return $Out_{Aud.}$

Algorithm 1. **MIAO** Framework

Loss function

During the training phase, our model's strategy involves optimizing three distinct objectives in tandem: two are designed for comprehension and the generation of one target. Each MIDI-audio pairing undergoes a single forward pass through the compute-intensive components of the MIDI encoder, audio encoder, and audio decoder. This phase is crucial as different functionalities are activated specifically to compute the trio of losses, thereby enhancing the efficiency of the computational workflow.

MIDI-audio contrastive loss (MAC)plays a critical role in training by aligning the feature spaces of MIDI and audio data, specifically focusing on synchronizing Out_{mtok} (MIDI token output) and Out_{atok} (audio token output). This alignment is essential for the model to differentiate positive MIDI-audio pairs from negative ones effectively, enabling a more nuanced understanding of musical relationships[47]. MAC leverages the ITC loss function[48], incorporating a momentum encoder to create feature representations that act as adaptive, soft-label training targets. These targets dynamically adapt to distinguish between positive and negative pairs by identifying potential positive instances within negative samples in Eqs. 7 and 8, thereby improving the model's ability to accurately capture complex audio and MIDI correlations.

$$L_{\mathrm{MAC}} = -\log \frac{\exp(\mathrm{sim}(i,t)/\tau)}{\sum_{t' \in T} \exp(\mathrm{sim}(i,t')/\tau)} \tag{7}$$

$$\mathrm{sim}(i,t) = \cos(\theta_{it}) = \frac{i \cdot t}{\|i\| \cdot \|t\|} = \frac{\sum_{k=1}^{d} i_k t_k}{\sqrt{\sum_{k=1}^{d} i_k^2}\sqrt{\sum_{k=1}^{d} t_k^2}} \tag{8}$$

where $sim(i, t)$ denotes the similarity between MIDI representation i and audio embedding t, τ is a temperature parameter, and T is a set of audio pieces including the correct one and negatives. This approach ultimately enhances representational accuracy, equipping the model with a refined capability to synchronize MIDI and audio features in the learned space.

MIDI-audio matching loss (MAM) is crucial in honing the model's multimodal representations, ensuring a precise correspondence between Out_{mcls} and Out_{acls}. Functioning as a binary classifier, the MAM assesses whether MIDI-audio pairs are matched or unmatched, utilizing a dedicated linear layer known as the MAM head for this task. To sharpen the model's evaluative accuracy, a hard negative mining technique is applied in Eq. 9[49]. This method selectively targets and incorporates negative pairs that are most similar in contrast, presenting greater challenges for the model to distinguish during the training process. Such a methodical selection process for loss computation significantly enhances the model's capability to recognize and align MIDI-audio pairings effectively.

$$L_{\mathrm{MAM}} = -[y \log(p) + (1 - y) \log(1 - p)] \tag{9}$$

where p denotes the predicted probability by the model that the MIDI and audio are a correct match. The binary ground truth labels $y \in \{0, 1\}$ in Eq. 9 are determined by the paired structure of our training data. For each batch containing N aligned MIDI-audio pairs, we assign $y = 1$ when the MIDI sequence and audio clip originate from the same musical composition (positive pair), and $y = 0$ for artificially mismatched combinations (negative pairs). This labeling scheme is inherent to our dataset construction, where each MIDI file is explicitly paired with its corresponding audio rendering during preprocessing. The labels thus provide automatic supervision for the model to learn the correspondence between matched MIDI-audio pairs while rejecting incorrect matches, without requiring additional manual annotation.

Audio modeling loss (AM) is instrumental in enabling the audio decoder to adeptly generate audio from MIDI sequences. It employs a cross-entropy loss function to direct the model's predictions of audio sample probabilities in an autoregressive manner in Eq. 10[50]. The integration of a label smoothing parameter, α, in the loss computation helps broaden the model's generalization, thereby reducing the risk of overfitting. AM's strategic design enhances the model's ability to convert MIDI data into elaborate and high-fidelity audio waveforms.

$$L_{\mathrm{AM}} = -\sum_{i=1}^{N} \left[(1 - \alpha) \cdot y_i \log(p_i) + \frac{\alpha}{K} \log(p_i) \right] \tag{10}$$

where y_i is the ground truth label, and p_i is the predicted probability, K is the number of possible audio sample values.

Experiments

Datasets

Cerberus4 dataset[51] is specifically designed for the complex task of separating and transcribing musical mixtures that include a variety of polyphonic and percussive instruments. Comprising 2100 professionally synthesized mixtures, each piece in the Cerberus4 is accompanied by isolated sound sources and their corresponding MIDI data. To suit specific audio processing needs, the Cerberus4's audio is uniformly downsampled to 16 kHz. It features diverse instrument combinations, including piano, guitar, bass, drums, and strings, distributed across thousands of segments. This structure makes Cerberus4 an extensive resource for training and evaluating music separation and transcription models.

MAESTROv3 dataset[52] created in partnership with the International Piano-e-Competition, encompasses around 200 hours of high-quality, uncompressed audio. The audio in the MAESTROv3 is uncompressed, offering a high-resolution listening experience with a sample rate between 44.1 and 48 kHz and a 16-bit PCM stereo format. The dataset comprises MIDI recordings collected over ten annual iterations of the competition. The MAESTROv3 features virtuoso pianists performing on Yamaha Disklaviers, capturing detailed MIDI data such as key strike velocities and various pedal positions for nuanced musical expression. This precision, along with about 3 ms of audio-MIDI alignment, enables accurate remote judging of competitions. Each recording is meticulously segmented into individual pieces, annotated with the composer, title, and year of performance. The collection primarily focuses on classical repertoire, covering works from the 17th to early 20th century.

MusicNet dataset[53] is a comprehensive collection of classical music, designed for music research and machine learning applications. It includes hundreds of classical recordings from 10 composers, played on 11 instruments, and provides over one million temporal labels in 34 hours of music. The MusicNet consists of 330 classical recordings, each varying in length, and features a wide range of labels from 513 distinct instrument/note combinations. The recordings, sourced from various archives, are paired with meticulously aligned digital MIDI scores to ensure precise labeling.

Slakh2100 dataset[54] is meticulously designed to enhance music source separation and multi-instrument automatic transcription, with a focus on its synthetic multi-instrument characteristics. It features a comprehensive collection of multi-track audio and precisely aligned MIDI files, synthesized from 187 distinct patches across 34 classes using high-quality, sample-based virtual instruments. This synthetic approach results in 2100 unique, automatically mixed tracks that blend various musical elements, providing a rich resource for exploring and analyzing complex musical compositions and instrumental interactions.

Guitarset dataset[55] offers a collection of 360 acoustic guitar excerpts, each approximately 30 seconds in length, recorded using hexaphonic pickups and Neumann U-87 microphones. These excerpts are performed by 6 players across 5 musical styles, 3 chord progressions, and 2 tempi, yielding a rich variety of comping and

soloing samples. Accompanying the recordings are comprehensive annotations, including pitch contours, MIDI notes, beat positions, and chords, facilitating a wide range of musicological and signal-processing research.

URMP dataset[56] provides a rich collection of 44 classical chamber music arrangements, spanning 28 unique works by 19 composers, broken down into 11 duets, 12 trios, 14 quartets, and 7 quintets. This comprehensive dataset includes not only audio and video recordings but also musical scores and detailed frame-level and note-level transcriptions, covering a wide range of simple to expressive pieces lasting from 40 seconds to 4.5 minutes. It's an invaluable asset for multi-modal music analysis, offering varied classical compositions across different instrumental arrangements.

Experiment setup

The development of the MIAO model utilizes a sophisticated setup on the PyTorch platform, supported by the computational capabilities of four NVIDIA A100 GPUs, each managing a substantial batch size of 1024. The AdamW optimization algorithm is the chosen method for refining the model parameters, which is further optimized by integrating a cosine learning rate scheduler, facilitating an effective ramp-up in the initial stages of training. The model configuration includes a temperature parameter (τ) set at 99 and a label smoothing parameter (α) at 0.13, starting with a learning rate of 0.001 and incorporating a weight decay of 0.05 to optimize training efficacy. This setup ensures a detailed and systematic approach to training, aiming to maximize the performance and efficiency of the MIAO model.

Evaluation metrics

The **MT3 Transcription** (MT3 T.) metric[57] evaluates the accuracy of a synthesis model in replicating specific notes and instruments. This technique involves processing the model's output through an MT3 transcription system to compute an F1 score, using the "Full" metric from the MT3 study and analyzed with mir_eval. For precise transcription, a note must align within ±50 ms of the intended onset, its offset should be within 0.2 times the reference duration or at least 50 ms, and it must exactly match the instrument program number from the input.

The **Fréchet Audio Distance** (FAD)[58] is a metric for assessing the quality of generated audio. It measures the similarity between the distribution of features extracted from real audio samples and those from synthesized audio. This is done by applying a deep learning model to extract features from both sets and then computing the Fréchet distance, a measure of similarity between the two feature distributions. A lower FAD score indicates a closer resemblance of the synthesized audio to real audio, implying a higher quality of the generated samples.

The **Scale Consistency** (SC)[59] in the context of music objective metrics generally refers to the degree to which a piece of music maintains its characteristic scale or key throughout a performance or a generated piece. It is a measure of the accuracy and stability of the tonal center, ensuring that the notes produced align with the expected scales and key signatures, which are foundational elements of musical theory and composition. This metric is particularly relevant in the evaluation of music generation systems, where maintaining consistent scale is crucial for the coherence and listenability of the music.

The **Pitch Class Entropy** (PCE)[60] is an objective metric used in music analysis that quantifies the unpredictability or complexity of the distribution of pitch classes (the set of all pitches that are a whole number of octaves apart) in a piece of music. It is calculated using the concept of entropy from information theory, which, in this context, measures the amount of uncertainty or surprise in the occurrence of different pitch classes. A higher Pitch Class Entropy value would indicate a more varied and unpredictable use of pitches, suggesting a complex piece, while a lower value would suggest more repetition and predictability in the use of pitches.

The **Reconstruction Embedding Distance** (RED)[43] metric evaluates the similarity between an original audio clip and its synthetic counterpart by analyzing them through a classifier network. It measures the divergence between the signals using the network's embedding space, employing the Frobenius norm across time frames for distance calculation. This metric is assessed using two architectures: VGGish, which generates one embedding per second of audio using its output layer, and TRILL, which produces approximately 5.9 embeddings per second from a specialized embedding layer.

Performance

MIAO is engineered to deliver high-fidelity audio synthesis, encompassing a broad spectrum of instruments and vocal ranges. It stands out as a holistic end-to-end solution capable of directly converting MIDI sequences into nuanced audio, fostering a highly interactive and expressive synthesis environment. Empirical evaluation in Table 1 underscores MIAO's capabilities, where it is rigorously benchmarked against six datasets: MAESTROv3, Slakh2100, Cerberus4, Guitarset, MusicNet, and URMP, each presenting a unique set of challenges in terms of instrument complexity and performance dynamics.

On the Cerberus4 and MAESTROv3 datasets, MIAO shows a measurable improvement over other models across multiple metrics. For instance, MIAO achieves 0.52 of MT3 T., approximately 67% higher on Cerberus4 compared to Midi2Wave[18], demonstrating its precision in reproducing musical details. Additionally, MIAO's RED score is around 26% lower than MI-DDPM, suggesting a closer match to the original audio. On MAESTROv3, MIAO's FAD is about 40% lower than that of MIDI-DDSP[61], reflecting its ability to generate audio distributions that closely resemble real samples, enhancing the quality of synthesized music.

For the MusicNet and Slakh2100 datasets, MIAO maintains a distinct advantage in terms of MT3 T and audio fidelity. MIAO achieves a score of 0.28 MT3 T. over 55% higher on MusicNet than Midi2Wave and reduces RED by 0.44 compared to MI-DDPM, showing its effective replication of musical features. On Slakh2100, MIAO's FAD score is 33% lower than MIDI-DDSP, indicating more realistic audio synthesis. Additionally, MIAO's higher PCE scores, around 11% greater than MIDI-DDSP, reflect its capacity to utilize a broader range of pitch classes, contributing to a richer and more intricate musical output.

Datasets	Models	MT3 T.↑	RED↓	FAD↓	SC↑	[illegible]↑
Cerberus4	Midi2Wave	0.32	2.37	0.43	0.75	2.13
	MIDI-DDSP	0.39	1.82	0.34	0.88	2.49
	MI-DDPM	0.44	1.73	0.22	0.87	2.84
	MIAO	0.52	1.35	0.18	0.93	2.97
MAESTROv3	Midi2Wave	0.19	3.38	0.52	0.69	2.12
	MIDI-DDSP	0.26	2.83	0.47	0.87	2.44
	MI-DDPM	0.30	2.39	0.35	0.92	2.57
	MIAO	0.35	2.16	0.28	0.93	2.82
MusicNet	Midi2Wave	0.18	4.31	0.46	0.82	2.37
	MIDI-DDSP	0.23	3.91	0.39	0.88	2.45
	MI-DDPM	0.25	3.29	0.33	0.91	2.84
	MIAO	0.28	2.85	2.852.85	0.93	2.93
Slakh2100	Midi2Wave	0.31	1.67	0.39	0.89	1.95
	MIDI-DDSP	0.33	1.57	0.29	0.91	2.11
	MI-DDPM	0.39	1.41	0.25	0.95	2.05
	MIAO	0.42	1.22	0.20	0.93	2.30
Guitarset	Midi2Wave	0.44	4.81	0.43	0.78	2.13
	MIDI-DDSP	0.48	4.44	0.37	0.87	2.19
	MI-DDPM	0.52	4.09	0.26	0.92	2.33
	MIAO	0.56	3.86	0.15	0.94	2.59
URMP	Midi2Wave	0.21	4.03	0.45	0.87	2.57
	MIDI-DDSP	0.25	3.75	0.37	0.95	2.64
	MI-DDPM	0.27	3.54	0.23	0.96	2.86
	MIAO	0.31	3.26	0.18	0.95	2.78

Table 1. Comprehensive model experimental results on multiple data sets.

Setting	MT3 T.↑	RED↓	FAD↓	SC↑	PCE↑
w/o AD	0.31	1.86	0.25	0.75	2.62
w AD	0.52	1.35	0.18	0.93	2.97

Table 2. Comparative performance analysis of MIAO with (w) and without (w/o) Audio Encoder on the Cerberus4 Dataset. Significant values are in bold.

On the Guitarset and URMP datasets, MIAO continues to achieve higher MT3 T. and SC. For Guitarset, MIAO's MT3 T. is approximately 27% greater than MI-DDPM, highlighting its precision in note representation. Its FAD on Guitarset is nearly 60% lower than Midi2Wave, suggesting a closer alignment with natural sound characteristics. On URMP, MIAO maintains the second-best scale consistency with a score of 0.95, comparable to MI-DDPM, while achieving an FAD score that is around 20% lower, indicating its ability to produce structured and harmoniously consistent audio across complex instrumentation. These results underline MIAO's adaptability in handling diverse music datasets and generating expressive audio.

In conclusion, the empirical data from Table 1 suggests that the MIAO model generally outperforms Midi2Wave, MIDI-DDSP, and MI-DDPM in transcription accuracy, audio quality, and pitch utilization complexity. MIAO's consistently lower FAD scores across datasets underscore its exceptional audio synthesis quality, demonstrating its ability to produce audio that closely resembles real samples, which is critical for achieving high harmony in synthesized music. While SC does not differentiate the models, indicating that all models maintain a similar level of scale consistency, MIAO's PCE performance suggests it strikes a balance between complexity and predictability in pitch class distribution, enhancing the richness and intricacy of the musical output. The higher PCE scores reveal MIAO's proficiency in utilizing a diverse range of pitch classes, contributing to more complex and expressive musical compositions. These findings indicate MIAO's robust capabilities as a synthesis model, particularly advantageous in applications requiring high harmony and intricate transcription in music audio generation. The model's ability to handle a wide spectrum of instruments and voices with precise note-level control, along with its performance across various challenging datasets, highlights its versatility and effectiveness. This positions MIAO as a significant advancement in the field of computational musicology, capable of meeting the nuanced demands of multi-instrument music synthesis and paving the way for more interactive and expressive music synthesis applications in the future.

Setting	MT3 T.↑	RED↓	FAD↓	SC↑	PCE↑
w/o AD	0.48	4.36	0.25	0.89	2.37
w AD	0.56	3.86	0.15	0.94	2.59

Table 3. Comparative performance analysis of MIAO with (w) and without(w/o) audio encoder on the guitarset dataset. Significant values are in bold.

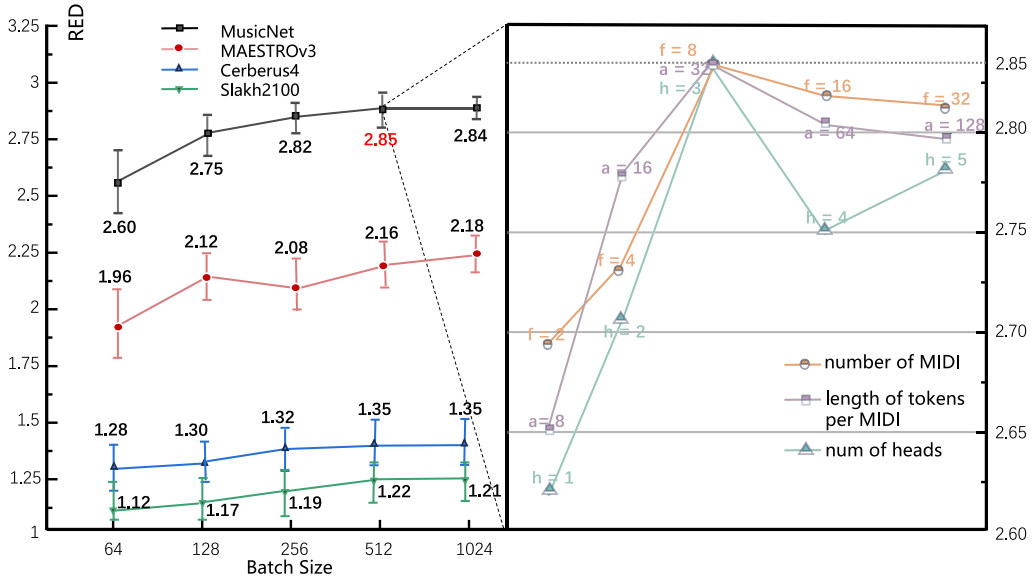

Fig. 2. The batch size (left) and parameter (right) settings.

Ablation study

Decoder

Table 2 demonstrates the performance impact of MIAO's audio decoder architecture. The first row shows the baseline results when replacing the proposed decoder with a simpler feed-forward network (FFN), while subsequent rows highlight how the full decoder configuration significantly improves performance across all metrics on the Cerberus4 dataset. This comparative analysis confirms that the structural components of our proposed audio decoder are essential for achieving state-of-the-art results. The MIAO's accuracy in note and instrument replication, as indicated by the MT3 Transcription score, improves by approximately 67.74%, rising from 0.31 to 0.52 with the audio decoder. The Reconstruction Embedding Distance metric decreases by about 27.42%, falling from 1.86 to 1.35, which denotes a closer resemblance of the synthesized audio to the original. Additionally, the Fréchet Audio Distance is reduced by 28%, from 0.25 to 0.18, highlighting the improved quality of the generated audio. Scale Consistency increases by 24%, from 0.75 to 0.93, suggesting that the audio decoder contributes to a more consistent maintenance of the music's scale. Finally, the Pitch Class Entropy sees a rise of 13.36%, from 2.62 to 2.97, indicating that the audio decoder aids in producing a richer and more complex pitch distribution. Overall, these improvements underscore the audio decoder's vital role in advancing the MIAO model's multi-instrument music synthesis performance.

Encoder

Table 3 illustrates the impact of the audio encoder on MIAO's performance on the Guitarset Dataset. With the audio encoder, the MT3 Transcription score improves by 16.67% from 0.48 to 0.56, indicating better transcription accuracy. The Reconstruction Embedding Distance is reduced by approximately 11.47%, from 4.36 to 3.86, suggesting more accurate audio synthesis. The Fréchet Audio Distance sees a substantial decrease of 40%, from 0.25 to 0.15, reflecting an improvement in the audio quality. Scale Consistency increases by 5.62%, from 0.89 to 0.94, showing a slight enhancement in maintaining the music's scale. Pitch Class Entropy rises by 9.28%, from 2.37 to 2.59, indicating a more diverse pitch class distribution. These metrics collectively signal the significant advantage of including an audio encoder in the MIAO model for the Guitarset Dataset, enhancing its ability to generate high-fidelity musical audio with detailed transcriptional intricacy.

Parameter setting

Figure 2 (right) delves into how specific parameter settings affect MIAO's performance on the MusicNet dataset, particularly at a batch size of 512. The examination highlights optimal parameter configurations for maximizing model accuracy. For the number of heads in the multi-head attention mechanism (h), performance increases

with the number of heads, achieving optimal RED at 2.85 for 'h = 3'. Adding more heads, such as 'h = 4' or 'h = 5', does not significantly enhance performance and may even slightly degrade it, suggesting that three heads optimally balance computational efficiency and capability for diverse feature representation. Similarly, the length of tokens per MIDI (a) shows that 'a = 32' reaches the highest RED of 2.85, while extending the token length further slightly reduces accuracy. For the number of MIDI inputs (f), the optimal performance is observed at 'f = 8', where RED also peaks at 2.85; increasing 'f' beyond this point results in diminished returns, indicating an overload of input data that does not contribute to, and may even hinder, model performance. These results underscore the importance of calibrating these parameters within specific bounds to avoid over-complication and ensure efficient model operation.

Limitations and future plans

MIAO demonstrates strong applicability in multi-instrument music synthesis, achieving precise note-level control across diverse instruments, as evidenced by its performance on published music datasets. However, its current evaluation scope remains limited to harmonious compositions, leaving its efficacy on dissonant or non-harmonic music untested–a critical gap for broader adoption. To advance this framework, future work will focus on three key directions: (1) validating generalizability to unconventional musical structures, (2) integrating real-time interactive performance capabilities, and (3) incorporating emotion-aware synthesis to enhance expressive depth. These enhancements will solidify MIAO's role as a versatile and transformative tool in computational musicology.

Conclusion

In conclusion, the MIAO synthesizer emerges as a transformative entity in the realm of computational musicology, adeptly addressing the intricate demands of multi-instrument music synthesis. The integration of complex vision-language objectives within its innovative architecture enhances the synthesizer's ability to understand and replicate detailed musical structures. This capability significantly enriches the authenticity and expressiveness of the generated compositions, effectively bridging the often challenging gap between precision and versatility in MIDI to high-fidelity audio conversion. However, despite its advanced capabilities, MIAO still faces limitations in real-time processing speed and the handling of extremely complex acoustic environments.

Data availibility

All data is available, which can be requested on links: Cerberus4: https://interactiveaudiolab.github.io/demos/cerberus. MAESTROv3 : https://magenta.tensorflow.org/datasets/maestro. MusicNet: https://zenodo.org/records/5120004. Slakh2100: http://www.slakh.com/. Guitarset: https://zenodo.org/records/3371780. URMP: https://labsites.rochester.edu/air/projects/URMP.html

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References

1. Tur, A. O., Dall'Asen, N., Beyan, C. & Ricci, E. Exploring diffusion models for unsupervised video anomaly detection. In *2023 IEEE International Conference on Image Processing (ICIP)* 2540–2544 (IEEE, 2023).
2. Hayes, B., Shier, J., Fazekas, G., McPherson, A. & Saitis, C. A review of differentiable digital signal processing for music & speech synthesis. arXiv:2308.15422
3. Nakatsuka, T., Hamasaki, M. & Goto, M. Content-based music-image retrieval using self-and cross-modal feature embedding memory. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision* 2174–2184 (2023).
4. Pasquier, P. et al. MIDI-GPT: A controllable generative model for computer-assisted multitrack music composition. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 39, 1474–1482 (2025).
5. Li, J., Li, D., Xiong, C. & Hoi, S. BLIP: Bootstrapping language-image pre-training for unified vision-language understanding and generation. In *International Conference on Machine Learning* 12888–12900 (PMLR, 2022).
6. Wang, Z. et al. Toward learning joint inference tasks for IASS-MTS using dual attention memory with stochastic generative imputation. *IEEE Trans. Neural Netw. Learn. Syst.* https://doi.org/10.1109/TNNLS.2023.3305542 (2023).
7. Moliner, E., Lehtinen, J. & Välimäki, V. Solving audio inverse problems with a diffusion model. In *ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)* 1–5 (IEEE, 2023).
8. Mariani, G. et al. Multi-source diffusion models for simultaneous music generation and separation. arXiv:http://arxiv.org/abs/2302.02257arXiv:2302.02257 (2023).
9. Marougkas, A., Troussas, C., Krouska, A. & Sgouropoulou, C. Virtual reality in education: A review of learning theories, approaches and methodologies for the last decade. *Electronics* **12**, 2832 (2023).
10. Talwar, S., Kaur, P., Nunkoo, R. & Dhir, A. Digitalization and sustainability: Virtual reality tourism in a post pandemic world. *J. Sustain. Tour.* **31**, 2564–2591 (2023).
11. Taulli, T. The impact on major industries: A look at music, education, journalism, gaming, healthcare, and finance. In *Generative AI: How ChatGPT and Other AI Tools Will Revolutionize Business* 175–188 (Springer, 2023).
12. Lavengood, M. L. & Williams, E. The common cold: Using computational musicology to define the winter topic in video game music (RESUB). *Music Theory Online*, Vol. 29 (2023).
13. Pang, Y. et al. Slim UNETR: Scale hybrid transformers to efficient 3D medical image segmentation under limited computational resources. *IEEE Trans. Med. Imaging* **43**, 994–1005 (2023).
14. Pang, Y. et al. Automatic detection and quantification of hand movements toward development of an objective assessment of tremor and bradykinesia in parkinson's disease. *J. Neurosci. Methods* **333**, 108576 (2020).
15. Pang, Y. et al. Online self-distillation and self-modeling for 3D brain tumor segmentation. *IEEE J. Biomed. Health Inf.* https://doi.org/10.1109/JBHI.2025.3530715 (2025).
16. Kim, W., Son, B. & Kim, I. Vilt: Vision-and-language transformer without convolution or region supervision. In *International Conference on Machine Learning* 5583–5594 (PMLR, 2021).
17. Briot, J.-P. & Pachet, F. Deep learning for music generation: Challenges and directions. *Neural Comput. Appl.* **32**, 981–993 (2020).

18. Shi, X., Cooper, E., Wang, X., Yamagishi, J. & Narayanan, S. Can knowledge of end-to-end text-to-speech models improve neural midi-to-audio synthesis systems? In *ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)* 1–5 (IEEE, 2023).
19. Ji, S., Luo, J. & Yang, X. A comprehensive survey on deep music generation: Multi-level representations, algorithms, evaluations, and future directions. arXiv:http://arxiv.org/abs/2011.06801arXiv:2011.06801 (2020).
20. Zhang, D., Hu, Z., Li, X., Tie, Y. & Qi, L. Multi-track music generation network based on a hybrid learning module. In *2023 IEEE International Conference on Multimedia and Expo Workshops (ICMEW)* 326–331 (IEEE, 2023).
21. Renault, L. Differentiable piano model for midi-to-audio performance synthesis. In *JJCAAS 2023: Journées Jeunes Chercheurs en Audition, Acoustique musicale et Signal audio* (2023).
22. Renault, L., Mignot, R. & Roebel, A. Ddsp-piano: A neural sound synthesizer informed by instrument knowledge. *AES-J. Audio Eng. Soc. Audio-Accoustics-Appl.* **71**, 552–565 (2023).
23. Jonason, N. et al. DDSP-based neural waveform synthesis of polyphonic guitar performance from string-wise midi input. arXiv:http://arxiv.org/abs/2309.07658arXiv:2309.07658 (2023).
24. Yi, H. et al. Generating holistic 3D human motion from speech. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 469–480 (2023).
25. Kim, J., Oh, H., Kim, S., Tong, H. & Lee, S. A brand new dance partner: Music-conditioned pluralistic dancing controlled by multiple dance genres. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 3490–3500 (2022).
26. Zhu, P. et al. ERNIE-music: Text-to-waveform music generation with diffusion models. arXiv:http://arxiv.org/abs/2302.04456arXiv:2302.04456 (2023).
27. Lu, P. *et al.* MuseCoco: Generating symbolic music from text. arXiv:http://arxiv.org/abs/2306.00110arXiv:2306.00110 (2023).
28. Wang, Y., Chen, M. & Li, X. Continuous emotion-based image-to-music generation. *IEEE Trans. Multimedia* **26**, 5670–5679 (2023).
29. Li, J., Li, D., Savarese, S. & Hoi, S. BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. arXiv:http://arxiv.org/abs/2301.12597arXiv:2301.12597 (2023).
30. Li, S. & Sung, Y. Melodydiffusion: Chord-conditioned melody generation using a transformer-based diffusion model. *Mathematics* **11**, 1915 (2023).
31. Civit, M., Civit-Masot, J., Cuadrado, F. & Escalona, M. J. A systematic review of artificial intelligence-based music generation: Scope, applications, and future trends. *Expert Syst. Appl.* **209**, 118190 (2022).
32. Oord, A. v. d. et al. WaveNet: A generative model for raw audio. arXiv:http://arxiv.org/abs/1609.03499arXiv:1609.03499 (2016).
33. Mehri, S. et al. SampleRNN: An unconditional end-to-end neural audio generation model. arXiv:http://arxiv.org/abs/1612.07837arXiv:1612.07837 (2016).
34. Goodfellow, I. et al. Generative adversarial networks. *Commun. ACM* **63**, 139–144 (2020).
35. Donahue, C., McAuley, J. & Puckette, M. Adversarial audio synthesis. arXiv:http://arxiv.org/abs/1802.04208arXiv:1802.04208 (2018).
36. Engel, J. et al. GANSynth: Adversarial neural audio synthesis. arXiv:http://arxiv.org/abs/1902.08710arXiv:1902.08710 (2019).
37. Morrison, M. et al. Chunked autoregressive GAN for conditional waveform synthesis. arXiv:http://arxiv.org/abs/2110.10139arXiv:2110.10139 (2021).
38. Greshler, G., Shaham, T. & Michaeli, T. Catch-a-waveform: Learning to generate audio from a single short example. *Adv. Neural. Inf. Process. Syst.* **34**, 20916–20928 (2021).
39. Caillon, A. & Esling, P. Streamable neural audio synthesis with non-causal convolutions. arXiv:http://arxiv.org/abs/2204.07064arXiv:2204.07064 (2022).
40. Surís, D., Vondrick, C., Russell, B. & Salamon, J. It's time for artistic correspondence in music and video. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 10564–10574 (2022).
41. Wang, Y. et al. Tacotron: Towards end-to-end speech synthesis. arXiv:http://arxiv.org/abs/1703.10135arXiv:1703.10135 (2017).
42. Ho, J., Jain, A. & Abbeel, P. Denoising diffusion probabilistic models. *Adv. Neural. Inf. Process. Syst.* **33**, 6840–6851 (2020).
43. Hawthorne, C. et al. Multi-instrument music synthesis with spectrogram diffusion. arXiv:http://arxiv.org/abs/2206.05408arXiv:2206.05408 (2022).
44. Zhuo, L. et al. Video background music generation: Dataset, method and evaluation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 15637–15647 (2023).
45. Hendrycks, D. & Gimpel, K. Gaussian error linear units (GELUs). arXiv:http://arxiv.org/abs/1606.08415arXiv:1606.08415 (2016).
46. Khan, S. et al. Transformers in vision: A survey. *ACM Comput. Surv. (CSUR)* **54**, 1–41 (2022).
47. Radford, A. et al. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning* 8748–8763 (PMLR, 2021).
48. Li, J. et al. Align before fuse: Vision and language representation learning with momentum distillation. *Adv. Neural. Inf. Process. Syst.* **34**, 9694–9705 (2021).
49. Wang, Q., Gu, J.-C. & Ling, Z.-H. Multiscale matching driven by cross-modal similarity consistency for audio-text retrieval. In *ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)* 11581–11585 (IEEE, 2024).
50. Raffel, C. *Learning-Based Methods for Comparing Sequences, with Applications to Audio-to-Midi Alignment and Matching* (Columbia University, New York, 2016).
51. Manilow, E., Seetharaman, P. & Pardo, B. Simultaneous separation and transcription of mixtures with multiple polyphonic and percussive instruments. In *ICASSP 2020-2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)* 771–775 (IEEE, 2020).
52. Wu, Y. *et al.* MIDI-DDSP: Detailed control of musical performance via hierarchical modeling. arXiv:http://arxiv.org/abs/2112.09312arXiv:2112.09312 (2021).
53. Thickstun, J., Harchaoui, Z. & Kakade, S. Learning features of music from scratch. arXiv:http://arxiv.org/abs/1611.09827arXiv:1611.09827 (2016).
54. Manilow, E., Wichern, G., Seetharaman, P. & Le Roux, J. Cutting music source separation some Slakh: A dataset to study the impact of training data quality and quantity. In *2019 IEEE Workshop on Applications of Signal Processing to Audio and Acoustics (WASPAA)* 45–49 (IEEE, 2019).
55. Xi, Q., Bittner, R. M., Pauwels, J., Ye, X. & Bello, J. P. GuitarSet: A dataset for guitar transcription. In *ISMIR*, 453–460 (2018).
56. Li, B., Liu, X., Dinesh, K., Duan, Z. & Sharma, G. Creating a multitrack classical music performance dataset for multimodal music analysis: Challenges, insights, and applications. *IEEE Trans. Multimedia* **21**, 522–535 (2018).
57. Gardner, J., Simon, I., Manilow, E., Hawthorne, C. & Engel, J. Mt3: Multi-task multitrack music transcription. arXiv:http://arxiv.org/abs/2111.03017arXiv:2111.03017 (2021).
58. Kilgour, K., Zuluaga, M., Roblek, D. & Sharifi, M. Fréchet audio distance: A reference-free metric for evaluating music enhancement algorithms. In *INTERSPEECH*, 2350–2354 (2019).
59. Toh, R. K. H. & Sourin, A. Generation of music with dynamics using deep convolutional generative adversarial network. In *2021 International Conference on Cyberworlds (CW)* 137–140 (IEEE, 2021).
60. Yang, L.-C. & Lerch, A. On the evaluation of generative models in music. *Neural Comput. Appl.* **32**, 4773–4784 (2020).
61. Wu, Y. *et al.* MIDI-DDSP: detailed control of musical performance via hierarchical modeling. In *International Conference on Learning Representations* (2022).

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Declarations

Competing interests

The authors declare no competing interests.

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